

ROSHAN RAJ MAHATO

📞 +91-9204775882 | ✉ roshanrajmahato1234@gmail.com | 🔗 linkedin.com/roshanrajmahato | 🐙 github.com/Roshanrajmahato

INTERNSHIP

Medical Chatbot – Context-Aware Information Retrieval

YBI Foundation (Remote) Jun 2024 – Jul 2024

- Developed a **Medical Chatbot** leveraging **Llama-2-7B-Chat LLM** within a **Retrieval-Augmented Generation (RAG)** pipeline for context-aware medical information retrieval.
- Integrated **LangChain** to orchestrate generative AI workflows and connected **Pinecone Vector Database** for scalable embedding storage and fast semantic retrieval.
- Built a **Flask** web interface with modular backend architecture, ensuring maintainable and production-ready code structure.

POSITIONS OF RESPONSIBILITY

- **1st Place – Quiz-A-Thon 5.0** (Group 2), Coding Club and Technical Team – demonstrated strong analytical thinking and technical problem-solving under competition.
- **Event Management Volunteer – TECHXHIBIT-25** (Engineering Project Exhibition 6.0) – coordinated participant teams during project presentations and technical demonstrations.
- **Cultural Fest Event Management Team Member** – led stage coordination, crowd management, and successful execution of large-scale college events.

TECHNICAL SKILLS

Programming Languages

- **Python** – OOP, File Handling, NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn
- **C** – Arrays, Pointers, Strings, Structures, Unions

Machine Learning

- **Algorithms:** Linear Regression, Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, Naive Bayes, k-NN, AdaBoost
- **Unsupervised:** PCA, K-Means Clustering, Hierarchical Clustering
- **Techniques:** Hyperparameter Tuning, Feature Engineering, Cross-Validation

Deep Learning

- CNNs, RNNs, ANNs, Neural Networks
- Frameworks: **TensorFlow**, **PyTorch**, Keras

Generative AI and NLP

- **LangChain**, **RAG**, **LLMs**, OpenAI GPT-4, HuggingFace Transformers, **FAISS**

Data Analysis

- **Pandas** – Data Cleaning, EDA, Descriptive Statistics
- **Power BI** – DAX, Power Query, Interactive Dashboards
- **Microsoft Excel** – VLOOKUP, HLOOKUP, Pivot Tables

Databases

- **SQL** – CRUD Operations, Joins, Sub-queries, Normalization, **MySQL**, Oracle SQL

Web Development

- **Django**, Flask, HTML5, CSS3, JavaScript, Bootstrap

EDUCATION

Ramgarh Engineering College Jun 2021 – Jul 2025
B.E. – Computer Science and Engineering CGPA: **7.0**

D.A.V Public School, Moonidih, Dhanbad Jun 2018 – Jul 2020

Intermediate – Class XII **89%**

D.A.V Public School, Moonidih, Dhanbad April 2018
Matriculation – Class X **85%**

SOFT SKILLS

- **Problem Solving** and Analytical Thinking
- **Communication** and Presentation Skills
- **Team Collaboration** and Leadership
- Adaptability and Fast Learning

TOOLS

- **IDE and Code Editors:** Visual Studio Code, PyCharm
- **Database Management:** MySQL Workbench, SQL*Plus – data modeling, query optimization, administration
- **Version Control:** Git, GitHub, Bitbucket – collaborative development, CI/CD pipelines
- **MLOps and Cloud:** MLflow, Docker, AWS (EC2, ECR), GCP (Cloud Functions, Buckets)

PROJECTS

Student Performance Prediction Dec 2023
Python, Scikit-learn, Machine Learning, Regression, Classification

- Achieved **96% accuracy** forecasting student academic performance by building and deploying an end-to-end supervised machine learning model.
- Improved data quality by **33%** through missing value imputation, categorical encoding, and feature engineering.
- Benchmarked multiple classification and regression algorithms to select the optimal model for deployment.

Pneumonia Detection and Classification Sept 2022
Python, TensorFlow, Keras, VGG16, Transfer Learning, CNN

- Built a **CNN using VGG16 Transfer Learning** to classify chest X-ray images as Pneumonia or Normal, achieving **91% accuracy**.
- Applied image augmentation and preprocessing techniques to improve model robustness for real-world medical imaging diagnostics.

Legal AI Assistant 2024
Python, LangChain, FAISS, OpenAI GPT-4, HuggingFace

- Built a **RAG-based Legal AI Assistant** that parses PDF and DOCX contracts, performing **clause extraction**, **risk scoring (HIGH/MEDIUM/LOW)**, and conflict detection using **LangChain** and **GPT-4**.
- Implemented **FAISS vector store** with **HuggingFace embeddings** for semantic search, enabling grounded Q&A with mandatory section citations.
- Designed a modular, production-ready pipeline with dedicated components for summarization, risk analysis, and contradiction detection.